MTH205 Analysis in one variable

Tutorial 4 DATE: **31.01.2025**

(1) Determine the points of continuity of the following functions.

(a) $f(x) = \frac{x^2+2x+1}{x^2+1}$, for all $x \in \mathbb{R}$.

(b) $g(x) = \sqrt{x + \sqrt{x}}$, for all $x \geq 0$.

(c) $h(x) = \frac{\sqrt{1+|\sin(x)|}}{x}$, for all $x \in \mathbb{R} \setminus \{0\}$.

(d) $\varphi(x) = \cos(\sqrt{1+x^2})$, for all $x \in \mathbb{R}$.

Solution: (a) Since the polynomial x^2+2x+1 is continuous on $\mathbb{R}$ and the denominator polynomial $x^2+1 \neq 0$ for every $x \in \mathbb{R}$ and continuous on $\mathbb{R}$. Therefore, the rational function f is continuous on $\mathbb{R}$.

(b) Let $x \geq 0$ and $\{x_n\}$ be any sequence of non-negative numbers with $\{x_n\} \to x$, as $n \to \infty$. Then

$$g(x_n) = \sqrt{x_n + \sqrt{x_n}} \longrightarrow \sqrt{x + \sqrt{x}}, \text{ as } n \to \infty.$$

This shows that the function g is a continuous function for every $x \geq 0$.

(c) Clearly, the function h is continuous on $\mathbb{R} \setminus \{0\}$ as both numerator, denominator (non-zero) functions are continuous on $\mathbb{R} \setminus \{0\}$.

(d) Since the function $\sqrt{1+x^2}$ and $\cos(x)$ is continuous on $\mathbb{R}$, it follows that the composition these two functions $\cos(\sqrt{1+x^2})$ is continuous on $\mathbb{R}$.

(2) Let $A \subseteq \mathbb{R}$. Give an example of a function $f : A \to \mathbb{R}$ such that f is discontinuous at every point of A but $|f|$ is continuous on A.

Solution: Define $f : A \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x \in A \cap \mathbb{Q}; \\ -1 & \text{if } x \in A \cap \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

then f is discontinuous at every point of A whereas the function $|f|(x) := |f(x)| = 1$ is continuous on A.

(3) Let $f : \mathbb{R} \to \mathbb{R}$ be continuous on $\mathbb{R}$ satisfying $f\left(\frac{m}{2^n}\right) = 0$ for every $m \in \mathbb{Z}$, $n \in \mathbb{N}$. Then show that $f(x) = 0$ for all $x \in \mathbb{R}$.

Solution: Since the sequence $\left\{\frac{1}{2^n}\right\} \to 0$ as $n \to \infty$. For $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $\frac{1}{2^N} < \epsilon$. Let $r \in \mathbb{R}$ and define $m := \lfloor r2^N \rfloor$ then

$$m \leq r2^N < m+1.$$

That is, $\frac{m}{2^N} \le r < \frac{m}{2^N} + \frac{1}{2^N}$ and we get $\left|r - \frac{m}{2^N}\right| < \frac{1}{2^N} < \epsilon$. In other words, the sequence $\left\{\frac{\lfloor r2^n \rfloor}{2^n}\right\}$ converges to r. Since f is continuous and $f\left(\frac{m}{2^n}\right) = 0$ we conclude that $f(r) = 0$, for all $r \in \mathbb{R}$.

(4) Suppose $f : \mathbb{R} \to \mathbb{R}$ is a function such that $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. Prove that if f is continuous at some point $x_0 \in \mathbb{R}$ then f is continuous at every point of $\mathbb{R}$.

Solution: First, note that $f(rx) = r\, f(x)$ for all $r > 0$, $x \in \mathbb{R}$. It follows that $f(-x) = f(x - 2x) = f(x) + 2f(-x)$ and so, $f(-x) = -f(x)$. As a result,

$$f(0) = f(x-x) = f(x) - f(x) = 0.$$

Now we show that f is continuous on $\mathbb{R}$ if and only if f is continuous at $x = 0$. The forward implication is trivial. conversely, if $\{x_n\}$ converges x_0 then the sequence $\{x_n - x_0\}$ converges to 0 and hence $f(x_n - x_0) = f(x_n) - f(x_0) \to 0$. Therefore, $f(x_n) \to f(x_0)$ as $n \to \infty$.

(5) Let $f, g : \mathbb{R} \to \mathbb{R}$ be continuous at a point $x_0 \in \mathbb{R}$. If we define $h(x) := \sup\left\{f(x), g(x)\right\}$ and $e(x) := \inf\{f(x), g(x)\}$ for $x \in \mathbb{R}$ then show that

(a) $h(x) = \frac{1}{2}\Big(f(x) + g(x)\Big) + \frac{1}{2}\Big|f(x) - g(x)\Big|$, $\forall\, x \in \mathbb{R}$ and h is continuous at x_o.

(b) $e(x) = \frac{1}{2}\Big(f(x) + g(x)\Big) - \frac{1}{2}\Big|f(x) - g(x)\Big|$, $\forall\, x \in \mathbb{R}$ and e is continuous at x_o.

Solution: (a) If $f(x) \ge g(x)$ then

$$\frac{1}{2}\Big(f(x) + g(x)\Big) + \frac{1}{2}\Big|f(x) - g(x)\Big| = f(x) = \sup\left\{f(x), g(x)\right\} = h(x).$$

If $f(x) \le g(x)$ then

$$\begin{aligned}\frac{1}{2}\Big(f(x) + g(x)\Big) + \frac{1}{2}\Big|f(x) - g(x)\Big| &= \frac{1}{2}\Big(f(x) + g(x)\Big) + \frac{1}{2}\Big(g(x) - f(x)\Big)\\ &= g(x)\\ &= \sup\left\{f(x), g(x)\right\}\\ &= h(x).\end{aligned}$$

The expression of $h(x)$ is true. In addition, h is continuous since the functions f, g are continuous on $\mathbb{R}$.

(b) Similarly, the function $e(x) = \inf\{f(x), g(x)\} = \frac{1}{2}\Big(f(x) + g(x)\Big) - \frac{1}{2}\Big|f(x) - g(x)\Big|$ for all $x \in \mathbb{R}$ is continuous on $\mathbb{R}$.